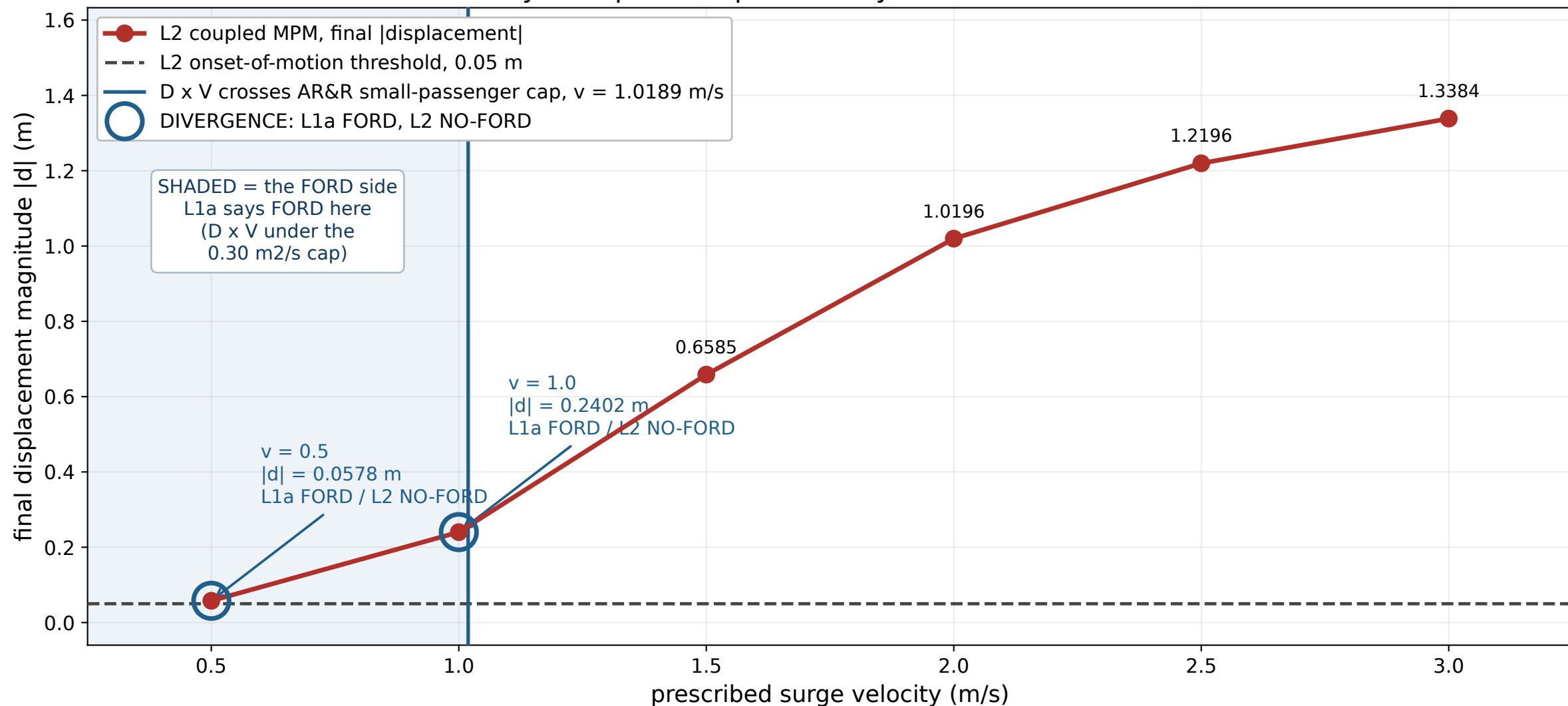


G1 Velocity sweep: the depth-velocity criterion fails on its own axis



Fixed grid 64, fixed realized depth 0.2944 m (4 layers x h), one Yaris hull, 1100 kg, all runs deterministic (determinism_identical = True).

THE ARGUMENT: the criterion is varied along the axis it is built on and still returns the wrong verdict at two of six points.

At $v = 0.5$ and $v = 1.0$ m/s the AR&R product D x V stays under the 0.30 m²/s small-passenger cap, so L1a returns FORD, while the coupled simulation moves the vehicle 0.0578 m and 0.2402 m, both above the 0.05 m onset threshold.

The vertical rule is where D x V = 0.30 m²/s at this fixed depth, $v = 1.018920$ m/s. AR&R limits are interim DRAFT criteria for stationary vehicles; "small passenger" names the limit set applied by kerb weight, not a measured vehicle class. On the AR&R geometric axes this hull satisfies only large_passenger (docs/four_rung_ladder.md:274-280).

Source: data/all_runs_inventory.csv, runs sweepV_g64_v0p5, sweepV_g64_v1p0, g64_m1100, sweepV_g64_v2p0, sweepV_g64_v2p5, sweepV_g64_v3p0.